SMS SPAM DETECTION

Problem Statement

SMS spam messages have become a significant issue in digital communication, causing inconvenience, security risks, and potential fraud for users. Detecting spam messages automatically is essential for improving communication security and filtering unwanted messages.

This project focuses on developing a machine learning-based SMS spam detection system that classifies SMS messages as either legitimate (ham) or spam. Using text classification techniques and Natural Language Processing (NLP), the project aims to improve spam detection accuracy and support safer communication systems.

Objectives:

- Classify SMS messages as spam or ham
- Apply text preprocessing and feature extraction techniques
- Compare multiple Naïve Bayes classifiers
- Evaluate performance using classification metrics
- Identify the most effective model for SMS spam detection

Dataset Used

Dataset Name:

SMS Spam Collection Dataset

Dataset Overview:

- Total Messages: 5,574 SMS messages
- Language: English
- Target Variable: Message Label (ham or spam)

Dataset Context:

The dataset was created for SMS spam research and contains labeled SMS text messages collected from publicly available research sources.

Data Format:

- Each row contains one SMS message

- Column v1: Label (ham or spam)
- Column v2: Raw SMS text

Dataset Sources:

- Grumbletext UK spam forum
- NUS SMS Corpus (National University of Singapore)
- Caroline Tag's PhD Thesis dataset
- SMS Spam Corpus v.0.1 Big

Data Characteristics:

- Structured text dataset
- Binary classification problem
- Text-based supervised learning

Approach / Methodology

Data Preparation:

- Loaded SMS Spam Collection dataset
- Used v2 column as text feature
- Used v1 column as target label
- Applied text preprocessing
- Converted text into numerical features using TF-IDF Vectorizer
- Split data into training and testing sets

Feature Extraction:

- **TF-IDF Vectorizer:**

  Term Frequency-Inverse Document Frequency (TF-IDF) was used to transform raw SMS text into numerical feature vectors suitable for machine learning algorithms.

Machine Learning Models Used:

- **Gaussian Naïve Bayes (GNB):**

  A probabilistic classifier based on Gaussian distribution assumptions.

- **Multinomial Naïve Bayes (MNB):**

  A classifier commonly used for text classification that works well with word count or frequency features.

- **Bernoulli Naïve Bayes (BNB):**

  A classifier suitable for binary feature vectors, particularly effective in spam detection.

Evaluation Metrics

The following performance metrics were used:

- Accuracy Score
- Confusion Matrix
- Precision Score

Results and Evaluation

Gaussian Naïve Bayes (GNB)

Accuracy Score:

86.46%

Confusion Matrix:

[[776, 119],
[21, 118]]

Precision Score:

49.79%

Performance Analysis:

- Lower accuracy compared to other models
- Higher false positives
- Moderate spam detection capability
- Lower precision indicates more legitimate messages incorrectly classified as spam

Multinomial Naïve Bayes (MNB)

Accuracy Score:

95.74%

Confusion Matrix:

[[895, 0],
[44, 95]]

Precision Score:

100%

Performance Analysis:

- High overall accuracy
- Perfect precision
- No false positives
- Very strong spam classification
- Some spam messages were missed

Bernoulli Naïve Bayes (BNB)

Accuracy Score:

96.13%

Confusion Matrix:

[[892, 3],
[37, 102]]

Precision Score:

97.14%

Performance Analysis:

- Highest overall accuracy
- Excellent precision
- Very low false positives
- Better spam detection than MNB
- Most balanced classifier

Comparative Analysis

Metric	Gaussian NB	Multinomial NB	Bernoulli NB
Accuracy Score	86.46%	95.74%	96.13%
Precision Score	49.79%	100%	97.14%
False Positives	119	0	3
False Negatives	21	44	37

Conclusion

This project successfully implemented machine learning models for SMS spam detection using TF-IDF feature extraction and Naïve Bayes classifiers.

Final Performance:

- Gaussian NB Accuracy: 86.46%
- Multinomial NB Accuracy: 95.74%
- Bernoulli NB Accuracy: 96.13%

Best Performing Model:

Bernoulli Naïve Bayes (BNB)

Reason:

Bernoulli NB provided:

- Highest accuracy
- Excellent precision
- Balanced spam detection
- Very low false positives
- Reliable overall classification performance

Thus, Bernoulli Naïve Bayes is the most suitable model for this SMS spam detection system.